MIT VNAV 2023 - Lab 2 Individual Solutions

ROS tf2, Relative Motion, and Quaternion Identities

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Deliverable 1: Nodes, Topics, and Launch Logic

The static scenario contains four application nodes (excluding /rosout): /av1broadcaster, /av2broadcaster, /plots_publisher_node, and /rviz.

Node	Publishes	Subscribes / role
/av1broadcaster	/tf_static	Publishes the fixed world -> av1 frame.
/av2broadcaster	/tf_static	Publishes the fixed world -> av2 frame.
/plots_publisher_node	/visuals	Its tf2 listener receives /tf and /tf_static.
/rviz	-	Receives /visuals, /tf, and /tf_static; /visuals contains the meshes.

The same static scenario can be started without roslaunch by using one terminal for each command:

```
roscore
rosrun tf2_ros static_transform_publisher 1 0 0 0 0 0 1 world av1 __name:=av1broadcaster
rosrun tf2_ros static_transform_publisher 0 0 1 0 0 0 1 world av2 __name:=av2broadcaster
rosrun two_drones_pkg plots_publisher_node
rosrun rviz rviz -d $(rospack find two_drones_pkg)/config/default.rviz
```

If static:=True is omitted, the argument defaults to false. The group protected by if="$(arg static)" is skipped, while frames_publisher_node, protected by unless, is launched. It continuously publishes moving transforms on /tf.

Deliverables 2 and 3: Completed ROS Package

The submitted two_drones_pkg contains both completed sources. A single timestamp is used for both transforms. With elapsed time t, the translations and AV1 yaw are

$$ {}^{w}o_{1} = [\cos t, \sin t, 0]^{T}, \quad {}^{w}o_{2} = [\sin t, 0, \cos 2t]^{T}, \quad {}^{w}R_{1} = R_{z}(t). $$

The trail update calls lookupTransform(ref_frame, dest_frame, ros::Time(0)), which returns the newest available pose of dest_frame expressed in ref_frame.

Deliverable 4: Relative Trajectory

1. AV2 trajectory in the world frame

Because $x = \sin t$, $y = 0$, and $\cos 2t = 1 - 2\sin^2 t$,

$$y = 0, \quad z = 1 - 2x^2, \quad -1 \le x \le 1.$$

> Result. AV2 moves on an arc of the parabola $z = 1 - 2x^2$ in the world x-z plane.

2. AV2 expressed in AV1

AV1 has yaw t, while AV2 has axes parallel to the world axes. The homogeneous transforms are

$$^{w}T_1 = [\, R_z(t) \quad {}^{w}o_1 \,;\, 0 \quad 1 \,], \quad {}^{w}T_2 = [\, I \quad {}^{w}o_2 \,;\, 0 \quad 1 \,].$$

Therefore

$$^{1}T_2 = (^{w}T_1)^{-1}\,{}^{w}T_2 = [\, R_z(-t) \quad R_z(-t)(^{w}o_2 - {}^{w}o_1) \,;\, 0 \quad 1 \,].$$

Multiplying the translation explicitly gives

$$^{1}o_2(t) = [\, \sin t \cos t - 1, \quad -\sin^2 t, \quad \cos 2t \,]^{T}$$

$$= [\, \tfrac{1}{2}\sin 2t - 1, \quad \tfrac{1}{2}(\cos 2t - 1), \quad \cos 2t \,]^{T}.$$

3. Plane of the relative trajectory

The last two components satisfy $z = \cos 2t = 2y + 1$. Hence

$$\Pi: z - 2y - 1 = 0, \quad \text{with normal } n = [0, -2, 1]^{T}.$$

4. Centered coordinates on the plane

Center the new frame at $p_c = (-1, -\tfrac{1}{2}, 0)^{T}$ and choose

$$e_{x'} = [1,0,0]^{T}, \quad e_{y'} = (1/\sqrt{5})[0,1,2]^{T}, \quad e_{z'} = (1/\sqrt{5})[0,-2,1]^{T}.$$

These vectors are orthonormal, and $e_{z'}$ is normal to Π. Projecting $^{1}o_2 - p_c$ onto them gives

$$x' = \tfrac{1}{2}\sin 2t, \quad y' = (\sqrt{5}/2)\cos 2t, \quad z' = 0.$$

5. Ellipse and semiaxes

$$x'^2/(1/2)^2 + y'^2/(\sqrt{5}/2)^2 = \sin^2 2t + \cos^2 2t = 1.$$

> Result. The relative trajectory is an ellipse centered at (-1,-1/2,0) in AV1, with semiaxis lengths 1/2 and √5/2.

Deliverable 5: Quaternion Identities

Write the scalar-last quaternion as $q = [q_v^T, q_4]^T$. The maps from the lecture are equivalently

$$\Omega_1(q) = [\, q_4 I + [q_v]_x \quad q_v \,;\, -q_v^T \quad q_4 \,],$$

$$\Omega_2(q) = [\, q_4 I - [q_v]_x \quad q_v \,;\, -q_v^T \quad q_4 \,].$$

1. Orthogonality

Use $[q_v]_x^T = -[q_v]_x$ and $[q_v]_x^2 = q_v q_v^T - ||q_v||^2 I$. Direct block multiplication cancels every off-diagonal block and gives

$$\Omega_i(q)^T \Omega_i(q) = (||q_v||^2 + q_4^2)\, I_4 = I_4, \quad i = 1,2.$$

A square matrix with $A^T A = I$ also satisfies $AA^T = I$. Intuitively, left or right multiplication by a unit quaternion preserves quaternion norm, so both matrix representations must be orthogonal.

2. Mapping q to the identity quaternion

$$\Omega_1(q)^T q = \Omega_2(q)^T q = [0,0,0,1]^T.$$

The vector part is $q_4 q_v - q_4 q_v = 0$, and the scalar part is $||q_v||^2 + q_4^2 = 1$. This is exactly $q^* \otimes q = q \otimes q^* = 1$.

3. Commutativity of the two linear operators

$\Omega_1(x)$ performs left multiplication by x, while $\Omega_2(y)$ performs right multiplication by y. For an arbitrary quaternion z, associativity gives

$$x \otimes (z \otimes y) = (x \otimes z) \otimes y.$$

$$\text{Therefore} \quad \Omega_1(x)\Omega_2(y) = \Omega_2(y)\Omega_1(x).$$

Because transposition changes right multiplication by y into right multiplication by its conjugate in this matrix form, the same associativity argument yields

$$\Omega_1(x)\Omega_2(y)^T = \Omega_2(y)^T \Omega_1(x).$$

Optional Deliverable 6: Intrinsic vs. Extrinsic Rotations

A rotation about a fixed/world axis premultiplies the current attitude, whereas a rotation about a moving/body axis postmultiplies it. Thus applying the extrinsic sequence R_0, R_1, R_2 produces

$$R_{ext} = R_2 R_1 R_0.$$

Applying the intrinsic sequence in reverse order, R_2, R_1, R_0, gives

$$R_{int} = I\, R_2 R_1 R_0 = R_{ext}.$$

> Result. Any extrinsic sequence is equivalent to the reverse-order intrinsic sequence.

Reference: MIT VNAV, Lab 2 Exercises (2023), https://vnav.mit.edu/labs_2023/lab2/exercises.html